

Detailed Curriculum

For your success

Topics

MODULE 1 Prompt Engineering and API Integration

- AI vs ML vs Deep Learning
- Introduction to Generative AI (GenAI)
- Open-source vs Closed-source AI Models
- GenAI Platforms Overview (OpenAI, Anthropic, Mistral, Stable Diffusion, Midjourney)
- Ethics & Responsible AI Usage
- Prompt Anatomy (Role, Instruction, Input, Output)
- Prompting Techniques (Zero-shot, Few-shot, Role-based Prompting)
- Advanced Prompting (CoT, ToT, ReAct, Self-Consistency)
- Handling AI Responses (JSON parsing, streaming, batch outputs, error handling)
- AI Guardrails & Output Validation
- Conversation Memory & Context Management
- Working with OpenAI/Gemini APIs (Chat, Embeddings, Function Calling)
- API Security, Rate Limits & Key Management
- Debugging & Evaluating GenAI Outputs (Accuracy, Relevance, Reliability)

Tools

Python

Hugging Face

Hugging Face API

LLM API

Postman

MODULE 2 Basics of NLP and Retrieval-Augmented Generation (RAG)

- Text Preprocessing & Tokenization
- Text Representation & Feature Engineering
- Named Entity Recognition (NER) & POS Tagging
- Embeddings, Vectorization & Semantic Search
- Chunking Strategies for LLM Applications
- RAG Fundamentals & Architecture
- Retrieval + Generation Workflow in RAG
- Data Handling, Chunking & Indexing
- Vector Databases & Vector Stores
- Dense vs Sparse Retrieval & Hybrid Search
- Working with Text/Image Embeddings in Vector DBs
- Building a Simple RAG Pipeline with LlamaIndex
- Query Rewriting, Multi-hop & Multi-query Retrieval
- Reducing Hallucinations & Context Optimization
- Advanced RAG Techniques (Self-RAG, Corrective RAG, Adaptive RAG, Contextual Retrieval, LOTR)
- Why Basic RAG Fails & Advanced Improvement Strategies
- RAG Evaluation & Fallback Mechanisms
- RAGAS Metrics (Faithfulness, Relevance, Context Recall & Precision)

LangChain

LlamaIndex

Hugging Face

Embeddings

FAISS

VectorDB

Pinecone

MODULE 3 AI Agents and Multi-Agent Systems

- Introduction to AI Agents & Agent Architectures
- Types of AI Agents (Reflex, Goal-based, Utility-based, Tool-based)
- Reactive Agents & Agent Design Patterns
- Tool Calling, Execution Flows & Prompting Strategies
- Multi-Agent Systems (MAS) & Collaborative AI
- Agent Coordination, Negotiation & Consensus Mechanisms
- Popular Agent Frameworks (LangChain, LlamaIndex, AutoGen, CrewAI, LangGraph)
- ReAct Agents & Tool-Using Agents
- LangGraph for Stateful Agent Orchestration
- Framework Comparison: Choosing the Right Agent Stack
- Planning, Reasoning & Tool Orchestration in Agents
- Hierarchical & Collaborative Multi-Agent Patterns
- Goal-Oriented Agent System Design
- Multi-Agent Reinforcement Learning (MARL) Basics
- MCP (Model Context Protocol): Servers, Clients, Tools & Resources
- A2A (Agent-to-Agent) Communication Protocols
- Agentic RAG: Autonomous Retrieval + Reasoning Pipelines
- Single-Agent vs Multi-Agent Agentic RAG
- Agent Debugging, Tracing & Monitoring with LangSmith
- Latency, Performance & Cost Optimization for AI Agents

LangChain

LangGraph

AutoGen

CrewAI

Hugging Face

LangSmith

MCP SDK

MODULE 4 Deployment and Project Integration

- Web API Fundamentals (HTTP Methods: GET, POST, PUT, DELETE)
- Building APIs with FastAPI & Pydantic
- API Documentation with Swagger/OpenAPI
- Deploying AI Apps (Render, Railway, Replit, Hugging Face Spaces)
- Dockerization for AI Applications
- CI/CD Pipelines with GitHub Actions
- Kubernetes (K8s) Basics for GenAI Scaling
- GenAI System Architecture Review (Agents, RAG, APIs)
- Designing Modular AI Systems & Microservices
- FastAPI for LLM Orchestration & LangChain Servers
- Frontend Integration with Gradio & Streamlit
- UI/UX Basics for LLM Applications
- API Security, Environment Variables & Rate Limiting
- Monitoring, Logging & Error Handling
- Introduction to LLMOps & Production AI Workflows
- Prompt Versioning, Tracing & Performance Monitoring
- LLMOps Platforms (Langfuse, MLflow, LangGraph)
- Observability with Prometheus & Grafana
- Live Deployment Demo & Project Showcase

LangChain

LangGraph

FastAPI

Streamlit

Langfuse

FAISS

Docker

GCP

LangServe

Kubernetes

MODULE 5 AI-Assisted Development with Agentic IDEs

- Set up and run AI coding agents across Terminal, IDEs, Cloud, and GitHub
- Choose and switch between leading AI models for different tasks
- Optimize performance with Low, Medium, and High reasoning modes
- Control agent permissions with Read-Only, Auto, and Full Access modes
- Master AI-powered terminals, commands, workflows, and planning tools
- Build persistent project memory using AGENTS.md, PLANS.md, and context files
- Connect external tools and services through MCP (Model Context Protocol)
- Create reusable Skills and Subagents for faster development
- Automate coding, testing, code reviews, CI/CD pipelines, and GitHub workflows
- Orchestrate multiple AI agents working together on complex projects
- Extend agents with Hooks, custom tools, and production-ready automations
- Understand AI limitations, hallucinations, security risks, and best practices
- Implement human-in-the-loop review, approval, and quality-control processes
- Deploy reliable, scalable, and secure AI-assisted development workflows

CodeX

Claude Code

Antigravity

MCP Server

Github

CI/CD

Optional Modules

MODULE 6 Foundations of Neural Networks and Transformers

- Machine Learning Foundations (Linear & Logistic Regression)
- Perceptrons & Multi-Layer Perceptrons (MLPs)
- Loss Functions & Activation Functions
- Gradient Descent & Backpropagation
- Introduction to Deep Neural Networks
- CNNs for Computer Vision Applications
- RNNs & LSTMs for Sequential Data
- Limitations of RNNs & Rise of Transformers
- Attention Mechanism (Query, Key, Value)
- Transformer Architecture Explained
- Pretraining vs Fine-tuning in LLMs
- Transfer Learning Concepts & Applications
- Introduction to GANs & Diffusion Models
- Foundations of Modern Image Generation Systems (DALL·E, Stable Diffusion, Midjourney)

Pytorch

Tensorflow

Keras

Huggingface

BertViz

MODULE 7 Fine-Tuning LLMs

- Introduction to Fine-Tuning & Domain Adaptation
- Fine-Tuning vs RAG: When to Use Each
- Hybrid Workflows: RAG + Fine-Tuning
- SLM vs LLM for Fine-Tuning
- Choosing Models (LLaMA, Mistral, Phi)
- Domain-Adaptive Pre-Training (DAPT)
- Dataset Preparation & Instruction Tuning
- Popular Datasets (Alpaca, DPO, Preference Datasets)
- Data Cleaning, Formatting & Labeling Best Practices
- PEFT Techniques (LoRA, QLoRA)
- Fine-Tuning Pipelines with Hugging Face & PEFT
- Trainer API, Configurations & Training Workflows
- Preference Optimization (DPO, ORPO, KTO)
- Quantization & Efficient Inference
- GGUF, AWQ & Model Compression Techniques
- Serving Models with vLLM & Ollama
- Evaluation Metrics (Accuracy, F1, BLEU, ROUGE)
- Fine-Tuning Performance Comparison & Benchmarking
- Cost Optimization & Resource Estimation for Training/Inference

Huggingface

Ollama

Datasets

vLLM

Transformers

Projects You Will Build

Module 1: Prompt Engineering and API Integration

1. AI-Powered Scam Detection System

Tackle spammy content in emails and SMS texts by building an advanced AI scam detector that restricts fraudulent attempts - helping customers avoid the possible risk by explaining them in layman's terms.

Skills Mastered: LLMs for Message Analysis, AI Text Classification, Intent Extraction, Fraud Pattern Detection.

2. AI-Powered Product Insight Generator

Turn messy customer reviews into sorted product insights by building an intelligent insight generator that analyzes reviews and generates summaries - helping customers make informed decisions.

Skills Mastered: Prompt Chaining, Structured Output Generation, Dynamic Summarization Techniques.

Module 2: Basics of NLP and Retrieval-Augmented Generation (RAG)

1. Resume-to-Job Semantic Search Engine

Build an advanced AI matching engine that functions on contexts and enables HRs to find relevant candidates for a job opening, even if the candidates don't have exact phrases in their resumes.

Skills Mastered: Embeddings, Vector Databases, MLOps & Privacy.

2. Smart Helpdesk Assistant

Build an intelligent AI chatbot that optimizes customer support for an organization by responding in the most logical and accurate manner to real-time queries - using internal data and documents of an organization.

Skills Mastered: Retrieval-Augmented Generation (RAG), Retrieval Pipelines, Prompt Engineering.

Module 3: AI Agents and Multi-Agent Systems

1. AI-Powered Finance and Inventory Management Assistant

Optimize financial operations and inventory control process of a business with the help of an advanced AI assistant - based on data-driven insights that enable businesses to make more informed decisions.

Skills Mastered: Agent-Based Workflows, Structured Memory, Reasoning Chains.

2. Custom Agentic AI Ecosystem

Enter the role of a GenAI architect, where you're no longer just building a single model - you're designing an entire ecosystem of intelligent agents that work collaboratively to solve dynamic, real-world problems.

Skills Mastered: Agentic AI, Workflow Optimization & Automation, Ecosystem Integration.

Module 4: Deployment and Project Integration

1. AI-Powered HealthCare Management System

Transform fragmented patient experiences into seamless healthcare journeys by creating an effective AI assistant that understands symptoms, recommends the right doctors, automates appointment scheduling, and generates contextual summaries.

Skills Mastered: Multi-Agent Coordination, Retrieval-Augmented Generation (RAG).

2. Smart Customer Support Chatbot System

Build a full-stack GenAI chatbot that can power enterprise customer support by retrieving real-time answers from documentation, handling complex queries, and triggering workflows - for B2B and B2C as well.

Skills Mastered: Prompting, RAG, Multi-Agent Systems, Deployment.